

FLOWERS ENCYCLOPEDIA

A comprehensive reference for ten flowers used in the Flower AI classification system

This encyclopedia covers ten flower species: Bougainvillea, Chrysanthemum, Cosmos, Giant Calotrope (*Calotropis gigantea*), Glebionis Coronaria (Crown Daisy), Hibiscus, Marigold, Nayantara (*Catharanthus roseus* / Periwinkle), Petunia, and Tagar (*Tabernaemontana divaricata* / Crepe Jasmine). Each entry includes distribution in India, blooming seasons, cultural significance, uses, cultivation notes, and key facts.

1. BOUGAINVILLEA

Bougainvillea spectabilis / *B. glabra*

Scientific Name	Bougainvillea spectabilis / B. glabra
Family	Nyctaginaceae
Common Names	Bougainvillea, Paper Flower, Bugarvilia
Native Origin	South America — Brazil, Peru, Ecuador
Colors	Pink, magenta, purple, red, orange, white, yellow, and bi-coloured varieties. The vivid colour comes from bracts (modified leaves), not petals.
Blooming	October to May in India (peak November to March). In cooler hill stations, flowering occurs mainly in spring. In tropical coastal areas such as Goa and Kerala, it can bloom nearly year-round.

Distribution in India

Widely cultivated across all of India, particularly in Maharashtra, Rajasthan, Tamil Nadu, Karnataka, Andhra Pradesh, West Bengal, and Goa. Extremely common in urban landscapes, gardens, and highway dividers.

Description

Bougainvillea is one of the most spectacular flowering plants in the tropical and subtropical world. It is a vigorous, thorny, woody climber or sprawling shrub belonging to the family Nyctaginaceae. The plant is renowned for its brilliantly coloured bracts — modified leaves that surround the tiny, inconspicuous, white or yellow true flowers at the centre. Many people mistakenly identify the bracts as the petals.

Uses

Ornamental: The primary use is aesthetic. It is used in landscape design, garden borders, trellises, bonsai, and hanging baskets.

Traditional medicine: In Brazil and Mexico, decoctions of the bracts are used to treat coughs, sore throats, and bronchitis. In India, the leaves are used in folk remedies for diabetes, stomach ailments, and liver complaints.

Environmental: The dense thorny growth makes excellent security hedges. The plant stabilises slopes and reduces soil erosion in landscaped areas.

Key Facts

- The colourful parts that most people call flowers are actually bracts (modified leaves). The true flowers are tiny, tubular, and usually white or pale yellow.
- There are over 300 named cultivars of Bougainvillea worldwide, differing in bract colour, leaf variegation, and growth habit.

- Bougainvillea was introduced to India in the 19th century and is now so widespread that many people consider it a native Indian plant.
 - It is the official flower of the city of Pune, Maharashtra, and is used extensively in Mumbai's coastal landscape.
 - In Singapore and Thailand, Bougainvillea is a national symbol and used extensively in public landscaping.
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2. CHRYSANTHEMUM

Chrysanthemum morifolium / Dendranthema grandiflorum

Scientific Name	Chrysanthemum morifolium / Dendranthema grandiflorum
Family	Asteraceae (Daisy family)
Common Names	Chrysanthemum, Mum, Shevanti (Marathi), Chandramallika (Hindi/Sanskrit), Sevanthi (Telugu/Kannada)
Native Origin	China and northeastern Europe; cultivated for over 2,500 years in East Asia
Colors	Yellow, white, orange, red, pink, bronze, purple, bi-coloured, and green. The yellow and orange forms are most common in Indian markets.
Blooming	September to January in India, with peak flowering during October and November coinciding with Diwali and Kali Puja. Chrysanthemums are short-day plants — they initiate flowering when nights become longer than a critical length (around 13 hours), which naturally occurs in autumn in India.

Distribution in India

Grown extensively across India, especially in West Bengal (Kolkata and surroundings), Maharashtra, Karnataka, Tamil Nadu, Andhra Pradesh, and Madhya Pradesh. Commercial cultivation is concentrated around Pune, Bengaluru, and Navsari. Kolkata's flower market at Mullick Ghat handles massive chrysanthemum volumes for the autumn festival season.

Description

Chrysanthemum is one of the most commercially important ornamental flowers in India and globally. It belongs to the Asteraceae family and is a herbaceous perennial or annual depending on climate. The plant produces flower heads composed of many individual florets — tubular disc florets at the centre and ray florets (the apparent petals) around the outside. Different cultivars show extraordinary variation in head shape: single, semi-double, double, pompon, spider, quill, and incurved forms all exist.

Uses

Cut flowers: India's chrysanthemum cut flower industry is worth several hundred crore rupees annually. Garlands and decorations: Used extensively in religious offerings, wedding decorations, and festive garlands. Herbal tea: Dried chrysanthemum flowers are brewed into a calming, antioxidant-rich tea, particularly consumed across East and Southeast Asia. Traditional medicine: Used in Ayurveda for eye complaints, headaches, and fevers. Insecticide: Pyrethrin, a natural insecticide, is extracted from certain Chrysanthemum species (Chrysanthemum cinerariifolium).

Key Facts

- The chrysanthemum is one of the Four Noble Plants in Chinese culture, along with plum, orchid, and bamboo.
 - It is Japan's national flower and appears on the Japanese Imperial Seal and on Japanese passports.
 - In France, chrysanthemums are associated with funerals and should not be given as a gift in casual settings.
 - Chrysanthemum tea is one of the most popular herbal teas in China and Vietnam, consumed for its cooling properties and mild sedative effect.
 - India exports chrysanthemums to the Netherlands, Japan, and the Middle East, making it a significant foreign exchange earner in the floriculture sector.
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3. COSMOS

Cosmos bipinnatus / *Cosmos sulphureus*

Scientific Name	Cosmos bipinnatus / Cosmos sulphureus
Family	Asteraceae (Daisy family)
Common Names	Cosmos, Mexican Aster, Garden Cosmos
Native Origin	Mexico, Central America, and northern South America
Colors	Pink, magenta, crimson, white, and bi-coloured (<i>Cosmos bipinnatus</i>); orange, yellow, and red (<i>Cosmos sulphureus</i>).
Blooming	October to February in plains of India. In hill stations, June to October. Cosmos germinates and blooms readily within 8 to 10 weeks of sowing, making it one of the fastest garden flowers to bloom from seed.

Distribution in India

Commonly grown as a garden annual across India, particularly in the cooler months. Found in domestic gardens, public parks, and roadside plantings in West Bengal, Himachal Pradesh, Uttarakhand, Maharashtra, and Karnataka. Thrives at altitudes between 1,000 and 3,000 metres in hill stations like Darjeeling, Ooty, Kodaikanal, Shimla, and Mussoorie.

Description

Cosmos is a genus of flowering plants in the sunflower family Asteraceae, comprising about 40 species. The most commonly grown garden species are *Cosmos bipinnatus*, which produces pink, white, and crimson daisy-like flowers with yellow centres, and *Cosmos sulphureus*, which produces yellow, orange, and red flowers. The name 'cosmos' comes from the Greek word for harmony and order, which reflects the orderly arrangement of petals.

Uses

Ornamental: Used extensively in garden beds, borders, and as cut flowers in floral arrangements. **Pollinator gardens:** An important nectar source for butterflies, particularly the common tiger butterfly, and for various bee species. **Green manure:** The plant biomass is used as green manure and composted organic matter in home gardens. **Companion planting:** Grown alongside vegetables to attract pollinators and pest-controlling beneficial insects.

Key Facts

- Cosmos was introduced to Europe from Mexico in the 18th century and quickly became one of the most popular garden flowers worldwide.
- The plant is a symbol of peace, tranquillity, and love in many cultures, and is given as a gift to represent these feelings.
- A single cosmos plant can produce hundreds of flowers in a single season.

- Cosmos sulphureus contains yellow-orange carotenoid pigments that have been studied for potential use as natural food colouring agents.
 - In Japan, there are entire fields of cosmos planted specifically as tourist attractions during the autumn season.
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4. GIANT CALOTROPE

Calotropis gigantea

Scientific Name	Calotropis gigantea
Family	Apocynaceae (Dogbane family)
Common Names	Giant Calotrope, Crown Flower, Madar (Hindi), Akanda (Bengali/Odia), Arka (Sanskrit), Erukku (Tamil/Malayalam), Jilledu (Telugu), Ekka (Kannada)
Native Origin	South Asia and Southeast Asia — India, Bangladesh, Sri Lanka, China, Malaysia, Indonesia
Colors	Pale purple to lilac with white. The five-petalled star-shaped corolla is white to pale purple with a purplish corona. There is also a white-flowered variety (Calotropis procera has similar white flowers).
Blooming	Year-round throughout India. Peak flowering occurs from September to February. In West Bengal, the plant blooms prolifically from October onwards, coinciding with the Hindu festival season when the flowers are in greatest demand for religious offerings.

Distribution in India

One of the most widespread wild shrubs in India, found across all states from sea level to about 1,000 metres altitude. Common in wastelands, roadsides, dry riverbeds, coastal dunes, and degraded forests. Particularly abundant in West Bengal, Odisha, Bihar, Uttar Pradesh, Rajasthan, Maharashtra, Tamil Nadu, and Andhra Pradesh. Practically absent only in the extreme northeastern hill states and high-altitude Himalayan zones.

Description

Calotropis gigantea, commonly called the Giant Calotrope or Crown Flower, is a large flowering shrub or small tree native to South and Southeast Asia. It is one of the most ecologically and culturally significant plants of the Indian subcontinent. The plant is characterised by large, thick, oval, grey-green leaves with a waxy surface that repels water. The stems and leaves contain a milky white latex that is toxic if ingested but has significant medicinal properties in traditional medicine systems.

Uses

Religious: Flowers are offered to Lord Shiva, Lord Ganesha, and Goddess Kali in Hindu worship. The flowers, leaves, and root bark are used in various religious rituals.

Traditional medicine (Ayurveda): The dried root bark is used for treating skin diseases, rheumatism, bronchitis, and digestive disorders. The latex is applied externally for skin infections, boils, and toothache. Flowers are used in the treatment of asthma and catarrh.

Fibre: The stem bark yields a strong bast fibre used for making ropes, fishing nets, and coarse textiles in rural India. The silky seed hair (similar to silk floss) has been used for stuffing pillows and as a

substitute for cotton.

Biocontrol: The latex contains bioactive compounds including calotropin, calactin, and cardenolides that are toxic to insects and have been studied as natural pesticides.

Key Facts

- *Calotropis gigantea* is the official flower of Queen Liliuokalani of Hawaii, and the Crown Flower is the traditional flower used in Hawaiian leis.
 - The plant can survive in highly saline coastal conditions where most plants fail — making it important for sand dune stabilisation.
 - The silky white fibres attached to the seeds were tested during World War II as a substitute for kapok in life vests and sleeping bag insulation.
 - In Ayurvedic medicine, the plant is classified as one of the most potent and toxic medicinal plants — small doses are therapeutic, large doses dangerous.
 - The plant is known to survive prolonged drought by shedding leaves and entering a semi-dormant state, resuming growth when rains arrive.
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5. GLEBIONIS CORONARIA

Glebionis coronaria (formerly *Chrysanthemum coronarium*)

Scientific Name	Glebionis coronaria (formerly Chrysanthemum coronarium)
Family	Asteraceae (Daisy family)
Common Names	Crown Daisy, Garland Chrysanthemum, Edible Chrysanthemum, Shungiku (Japanese), Tong Ho (Chinese), Chop Suey Greens, Kikuna
Native Origin	Mediterranean region — southern Europe and northern Africa; introduced and naturalized across Asia
Colors	Bright yellow, white, and bi-coloured (yellow outer ring with white inner zone). The flower heads are 3 to 6 cm in diameter with multiple ray florets surrounding a central disc.
Blooming	February to April in plains of India. November to March at higher altitudes in northeastern states and Himalayan foothills. It is an annual that completes its full lifecycle from seed to flower within approximately 10 to 14 weeks in cool conditions.

Distribution in India

Grown as a cool-season vegetable and ornamental in northern and northeastern India, particularly in West Bengal, Assam, Sikkim, and hill states. Also found in parts of Karnataka and Tamil Nadu. More commonly seen in vegetable markets in cities with large East Asian communities. Grown in Darjeeling district for local consumption.

Description

Glebionis coronaria, widely known as Crown Daisy or Garland Chrysanthemum, is a flowering annual plant native to the Mediterranean region that has been extensively cultivated across Asia for both its culinary and ornamental value. The plant was formerly classified as Chrysanthemum coronarium and is closely related to the ornamental garden chrysanthemum. The taxonomic name Glebionis coronaria has been in use since 2002 when the genus was revised.

Uses

Culinary vegetable: Young leaves and shoots are consumed across East and Southeast Asia in stir-fries, hotpots, noodle soups, and salads. The distinctive herbal aroma is appreciated in these cuisines.

Ornamental: Grown in winter and spring garden beds across temperate regions and hill stations for its bright yellow daisy flowers.

Traditional medicine: In Chinese traditional medicine, the plant is used as a cooling herb to reduce fever, calm the liver, and improve digestion.

Pollinator value: The open daisy flowers are excellent sources of nectar for bees and butterflies in early spring.

Key Facts

- The scientific name changed from *Chrysanthemum coronarium* to *Glebionis coronaria* in 2002 following a taxonomic revision, which is why it still appears under the old name in many older references.
 - In Japan, Shungiku is an essential ingredient in the traditional winter hot pot dish Shabu-shabu and in temple cuisine (Shojin-ryori).
 - The plant has been cultivated in China for over a thousand years and is mentioned in ancient Chinese agricultural texts.
 - Unlike the ornamental chrysanthemum, *Glebionis coronaria* can be grown easily from seed in a kitchen garden and harvested within two months.
 - The flowers are edible and are used as a garnish in Japanese bento boxes and salads.
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6. HIBISCUS

Hibiscus rosa-sinensis

Scientific Name	Hibiscus rosa-sinensis
Family	Malvaceae (Mallow family)
Common Names	Hibiscus, China Rose, Shoe Flower, Jaba (Bengali), Gudhal (Hindi), Dasaval (Tamil), Chembarathi (Malayalam), Mandara (Sanskrit)
Native Origin	Tropical Asia; exact origin uncertain — possibly China or India
Colors	Red, pink, orange, yellow, white, peach, and multicoloured hybrid varieties. The deep red single-flowered form is the most culturally significant in Indian Hindu practice.
Blooming	Year-round throughout tropical India. In regions with a cool winter (northern plains, hills), flowering may reduce between December and February. In Kerala, West Bengal, Tamil Nadu, and other tropical states, the plant produces flowers daily throughout the year. Each individual flower opens and closes within a single day.

Distribution in India

One of the most universally cultivated ornamental plants in India, found in virtually every state. Grown in home gardens, temple courtyards, roadsides, parks, and institutional landscapes from Kanyakumari to Kashmir. Particularly abundant in West Bengal, Kerala, Tamil Nadu, Andhra Pradesh, Maharashtra, and Goa. A sacred plant in Bengali Hinduism, where Jaba (red hibiscus) is offered daily to Goddess Kali and Goddess Durga.

Description

Hibiscus rosa-sinensis, commonly called the China Rose or simply Hibiscus, is one of the most widely grown ornamental shrubs in tropical and subtropical regions worldwide. The plant belongs to the family Malvaceae and produces spectacular trumpet-shaped flowers of 8 to 15 cm diameter, each lasting for a single day before wilting. The flowers are characterised by a prominent staminal column — a tube formed by fused stamens — that extends well beyond the petals, giving the flower its distinctive brush-like appearance.

Uses

Religious: Red hibiscus flowers are the most commonly used flowers in Bengali Hindu worship, offered daily to Goddess Kali. Used in puja (worship) rituals throughout India.

Hair care: Hibiscus leaves and flowers are ground into a paste used as a hair pack in Ayurvedic treatment. Products containing hibiscus extracts are widely sold commercially for hair strengthening, reducing hair fall, and promoting growth.

Medicinal: Used in Ayurveda for treating hair loss, dandruff, menstrual disorders, fever, and skin conditions. The flowers are used to prepare a paste applied to wounds.

Food and beverage: The calyx of *Hibiscus sabdariffa* (Roselle) is used to make the popular drink Agua de Jamaica in Mexico and Bissap in West Africa. In India, hibiscus roselle is grown in Andhra Pradesh and used for making jam, juice, and herbal tea.

Natural pH indicator: Hibiscus extract changes colour with pH (red in acid, green in base) and is used as a natural indicator in chemistry education.

Key Facts

- Each hibiscus flower opens fully for just one day — petals unfurl in the morning and drop by evening, making daily replenishment essential for religious offerings.
 - In Bengali Hindu practice, it is believed that offering 1,008 red hibiscus flowers to Goddess Kali on a specific occasion fulfils important spiritual merit.
 - The petals of *Hibiscus rosa-sinensis* are used in Asian countries as a shoe polish substitute — rubbing petals on shoes produces a shine similar to black polish, which is why it is called Shoe Flower.
 - The hibiscus is one of the few flowers that is equally used in religious, medicinal, and cosmetic applications across South Asia.
 - Hibiscus flowers contain mucilage — a gel-forming substance — that thickens preparations and is used in natural hair conditioning masks.
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7. MARIGOLD

Tagetes erecta (African Marigold) / *Tagetes patula* (French Marigold)

Scientific Name	Tagetes erecta (African Marigold) / Tagetes patula (French Marigold)
Family	Asteraceae (Daisy family)
Common Names	Marigold, African Marigold, Genda (Hindi/Bengali), Zendu (Marathi), Banthige (Kannada), Chamanthi (Telugu), Marikkolundu (Tamil)
Native Origin	Mexico and Central America
Colors	Brilliant orange, yellow, golden yellow, lemon yellow, cream, and bi-coloured orange-yellow. Orange is the predominant colour in Indian commercial production.
Blooming	October to March is the main season across most of India. A second crop blooms June to August in some regions. Planted at the beginning of September for Navratri-Dussehra-Diwali demand, and again in November for the wedding season. Short-duration varieties can be brought to flower within 45 to 60 days from transplanting.

Distribution in India

The single most important commercial flower crop in India. Cultivated in virtually every state, with the highest production in Karnataka, Andhra Pradesh, Telangana, Tamil Nadu, and Maharashtra. West Bengal is a major producer for the Kolkata market. Karnataka alone accounts for approximately 30 percent of India's total marigold production. The Pune and Bengaluru wholesale markets are among Asia's largest marigold trading centres.

Description

Marigold (*Tagetes*) is India's most commercially important cut flower and loose flower crop. It is an annual herbaceous plant with a strong, distinctive aromatic scent arising from terpene compounds in the leaves and stems. The flower heads are composed of multiple layers of ray florets arranged in a globular or semi-globular shape. In *Tagetes erecta* (the African or American Marigold), the flower heads are 5 to 10 cm in diameter and produced on long stems. In *Tagetes patula* (the French Marigold), flower heads are smaller, 3 to 5 cm, and produced in greater abundance on compact plants.

Uses

Religious and ceremonial: The primary use in India — garlands, decorations, loose flower offerings, and festive ornamentation. Used across all Hindu, Sikh, and some Muslim celebrations.

Natural pest repellent: Marigold roots secrete a compound (thiopene) that repels root-knot nematodes in soil. It is planted as a companion crop alongside tomatoes, chillies, and other vegetables in integrated pest management systems.

Natural food colourant and nutraceutical: Marigold oleoresin contains lutein and zeaxanthin, carotenoid pigments that are added to poultry feed and are sold as eye health supplements in human

nutrition products.

Medicinal: Used in folk medicine for wound healing, skin inflammation, and eye disorders. Extracts of *Tagetes* species have demonstrated antimicrobial and antifungal activity.

Insect repellent: The scent of marigold foliage repels aphids, whiteflies, and other garden pests when planted as a border crop.

Key Facts

- India is the world's largest consumer of marigold flowers, driven almost entirely by religious and festive demand.
 - During the Dussehra and Diwali season, a single marigold grower near Pune or Bengaluru can earn several lakh rupees in just 2 to 3 weeks.
 - Marigold leaves contain pyrethrin-like compounds and have been used as a traditional mosquito repellent in rural India — branches are burned to drive away insects.
 - The word 'marigold' is derived from 'Mary's Gold', a reference to the Virgin Mary in European Christian tradition.
 - The Aztecs of Mexico considered marigold a sacred flower and offered it to their solar deity. The Spanish brought it to Europe in the 16th century, from where it spread to India.
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8. NAYANTARA

Catharanthus roseus (formerly *Vinca rosea*)

Scientific Name	Catharanthus roseus (formerly Vinca rosea)
Family	Apocynaceae (Dogbane family)
Common Names	Nayantara (Bengali, meaning 'star of the eye'), Periwinkle, Sadabahar (Hindi, meaning 'ever-blooming'), Barmasi (Hindi), Vinca, Madagascar Periwinkle, Old Maid, Cape Periwinkle
Native Origin	Madagascar; naturalised throughout tropical regions worldwide
Colors	Pink, deep rose, white, lavender, pale lilac, red, and bi-coloured (white with pink or red eye centre). Single and double-flowered forms exist in cultivated varieties.
Blooming	Year-round throughout India without interruption. The name Sadabahar (ever-blooming) and Barmasi (twelve months) in Hindi reflect this continuous flowering habit. The plant does not have a dormant or non-flowering period in tropical India.

Distribution in India

One of the most ubiquitous wild and garden plants in India. Found growing along roadsides, in wastelands, on railway embankments, in garden borders, and in pots virtually everywhere in the country from sea level to about 1,200 metres altitude. Particularly abundant in West Bengal (where it is called Nayantara and is culturally loved), Odisha, Andhra Pradesh, Tamil Nadu, Maharashtra, Kerala, and Uttar Pradesh. Flowers abundantly in the Kolkata urban landscape year-round.

Description

Catharanthus roseus, called Nayantara in Bengali and Sadabahar in Hindi, is a perennial flowering plant that is simultaneously one of India's most beloved garden flowers, most important medicinal plants, and most successful wild plants. The plant is technically native to Madagascar but has naturalised so thoroughly across tropical regions that it is now considered a cosmopolitan tropical species.

Uses

Pharmaceutical — cancer treatment: The primary global significance. Vincristine and vinblastine extracted from the plant are essential chemotherapy drugs used worldwide for leukaemia and lymphoma treatment.

Traditional medicine: Used across tropical Asia and Latin America for treating diabetes, hypertension, fever, sore throat, and skin disorders.

Ornamental: Among the most popular bedding and pot plants in Indian urban gardens due to its continuous blooming habit, heat tolerance, and minimal care requirements.

Companion planting: The plant's alkaloids have mild repellent effects on certain garden insects.

Key Facts

- The discovery of vincristine and vinblastine from this plant is one of the most significant achievements in ethnobotanical drug discovery of the 20th century.
 - Before the development of vincristine, childhood acute lymphoblastic leukaemia had a survival rate of less than 10 percent. Today, with vincristine-based chemotherapy, the survival rate exceeds 90 percent.
 - The global revenue from vincristine and vinblastine drugs runs into hundreds of millions of dollars annually, yet the country of origin (Madagascar) has historically received little benefit from this discovery — a case study in biopiracy debates.
 - The plant can produce a new bloom cycle continuously throughout the year in tropical climates, making it one of very few plants that flower without any interruption.
 - In West Bengal, the flower is so commonly associated with simplicity and accessible beauty that it appears in the poetry of Rabindranath Tagore and other Bengali literary figures.
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9. PETUNIA

Petunia x atkinsiana (Petunia x hybrida)

Scientific Name	Petunia x atkinsiana (Petunia x hybrida)
Family	Solanaceae (Nightshade family)
Common Names	Petunia, Garden Petunia
Native Origin	South America — Argentina, Uruguay, and southern Brazil
Colors	Purple, violet, pink, red, white, yellow, coral, lavender, striped, and picotee (white-edged). The hybrid series available in India include Grandiflora (large flowers) and Multiflora (many smaller flowers).
Blooming	November to March in plains of India. In hill stations such as Shimla, Manali, and Darjeeling, the season extends from April to July. Petunias are at their best during the cool, dry months of the north Indian winter.

Distribution in India

Widely grown as a cool-season ornamental annual in India, particularly in West Bengal, Maharashtra, Punjab, Himachal Pradesh, Uttarakhand, Rajasthan (in winter), and Karnataka. A popular winter bedding plant in Indian public parks, roadsides, and private gardens. Thrives particularly well in Kolkata, Pune, Delhi, Bengaluru, and Jaipur gardens during the cool dry season.

Description

Petunia x atkinsiana, commonly simply called Petunia, is a popular cool-season annual or tender perennial belonging to the Solanaceae (nightshade) family. It is a complex hybrid derived from several South American *Petunia* species. The flowers are funnel-shaped with five fused petals, creating a wide-mouthed trumpet. The flower surface ranges from smooth to ruffled or frilled depending on the cultivar. Some varieties are fragrant, especially in the evenings.

Uses

Ornamental: The primary use. Used in garden beds, borders, hanging baskets, window boxes, containers, and pots as seasonal colour plants.

Aromatic: Several heirloom petunia varieties (particularly the Grandiflora types) produce a sweet fragrance that intensifies in the evening, making them useful near seating areas and windows.

Pest trap cropping: In integrated pest management, petunias can act as trap crops for thrips and aphids, drawing these pests away from vegetable crops nearby.

Key Facts

- The petunia belongs to the Solanaceae family — making it a relative of the tomato, potato, pepper, tobacco, and eggplant.

- The night-fragrant varieties of petunia are pollinated by moths in their native South American habitat.
 - Trailing petunia varieties called 'Wave Petunias' were developed in Japan in the 1990s and became an international sensation for their ability to spread without needing deadheading.
 - Petunia research has contributed significantly to our understanding of plant genetics — the Petunia genome has been widely used in studies of flower colour, pattern formation, and scent biochemistry.
 - The name Petunia comes from the French word 'petun', derived from a Brazilian Tupi-Guarani word for tobacco — reflecting its family relationship with the tobacco plant.
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10. TAGAR

Tabernaemontana divaricata (formerly *Ervatamia coronaria*)

Scientific Name	Tabernaemontana divaricata (formerly Ervatamia coronaria)
Family	Apocynaceae (Dogbane family)
Common Names	Tagar (Bengali), Chandni (Hindi), Crepe Jasmine, Pinwheel Flower, East India Rosebay, Nero's Crown, Adam's Apple, Nandiar Vattai (Tamil), Nandyarvattom (Malayalam), Tagar (Sanskrit)
Native Origin	India, Bangladesh, Myanmar, and possibly southern China; widely cultivated across tropical Asia
Colors	Pure white. The single-flowered form has 5 pure white propeller-like twisted petals. A double-flowered cultivar (Tabernaemontana divaricata 'Flore Pleno') exists with multiple whorls of white petals resembling a small gardenia or rose.
Blooming	Year-round throughout tropical India. Peak flowering occurs during March to November. In West Bengal, the plant blooms most profusely from March through October, coinciding with the warm humid season. Even in relatively dry winters, some flowering continues.

Distribution in India

Extremely common throughout India, particularly in West Bengal, Odisha, Bihar, Uttar Pradesh, Kerala, Tamil Nadu, Andhra Pradesh, and Maharashtra. In West Bengal, it is one of the most common flowering shrubs in both urban and rural areas. Grown in home gardens, temple surroundings, parks, and roadside plantings. Particularly common in the Gangetic plains and coastal South India.

Description

Tabernaemontana divaricata, known as Tagar in Bengali and Chandni (moonlight) in Hindi, is one of the most familiar and culturally important flowering shrubs of the Indian subcontinent. The plant is a member of the Apocynaceae family and is closely related to the Giant Calotrope and Nayanantara. Like them, it produces a milky latex when cut and contains alkaloids that give it medicinal significance.

Uses

Religious: Among the most widely used flowers in Hindu worship across eastern and southern India. Offered to Shiva, Vishnu, Lakshmi, and other deities. Used in garlands for festivals and puja.

Ornamental: A popular hedging and border plant in Indian gardens, frequently used to create informal fragrant hedges.

Traditional medicine: Root bark extracts are used in Ayurveda and folk medicine for treating inflammation, fever, eye disorders, and skin diseases. The plant contains alkaloids that inhibit certain cancer cell lines in laboratory studies.

Fragrance: The evening fragrance of the flowers makes them suitable for planting near windows, verandahs, and relaxation areas. Fragrant oil from the flowers is used in traditional perfumery.

Landscaping: Due to its year-round foliage, manageable size, and continuous flowering, it is a standard choice for low maintenance landscape planting in Indian institutional and public spaces.

Key Facts

- The species name 'divaricata' means 'spreading widely' in Latin, referring to the spreading branching habit of the plant.
 - In West Bengal, Tagar has a strong association with Bengali cultural identity and appears in folk songs, literature, and as a symbol of simple, enduring beauty.
 - The double-flowered Chandni variety is sterile and does not produce fruit — it is maintained entirely by vegetative (cutting) propagation.
 - *Tabernaemontana* species are distributed across tropical Africa, Asia, and the Americas and are a rich source of monoterpene indole alkaloids that pharmaceutical researchers continue to study.
 - The genus *Tabernaemontana* is named after Jacob Theodor von Bergzabern, a 16th-century German botanist whose Latin name was *Tabernaemontanus*.
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